OVERLAND PARK
KANSAS



PUBLIC WORKS DEPARTMENT

March 1, 2024

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1. GENERAL REQUIREMENTS

As stated in Chapter 13.12 of the Overland Park Municipal Code ("OPMC"), all earth, materials, sidewalk, pavement, crossing, storm sewer, utilities, public improvements or improvements of any kind damaged or removed by the service provider shall be fully repaired and replaced promptly by the ROW-user at its sole expense and the reasonable satisfaction of the City. Upon determination by the City Engineer that such repair or replacement is a public safety matter, all such repair or replacement shall be commenced within 24 hours of notice from the City, or the City Engineer may direct the City to make such repair or replacement and charge the ROW-user for the City's cost.

All construction work and materials shall conform to the current edition of the *Design and Construction Standards, Volume 2, Construction Specifications* (hereinafter "*Construction Specifications*"), unless otherwise directed by the City Engineer. *(See OPMC 13.10.010 C & L)*

All installation, construction, excavation, backfilling, restoration and replacement work shall be in accordance with:

- The current *Standard Details,* adopted and amended by the City Engineer. *(See OPMC 13.10.010 L & M)*; and
- The current Approved Materials List adopted and amended by the City Engineer.

After any excavation, the permittee shall restore all portions of the Right-of-way to the same condition or better condition that it was prior to the excavation. The permittee shall provide high quality color digital photographs and/or audio/video recording of the Right-of-way surface conditions prior to and after the completion of any excavation activities.

All asphaltic concrete shall be constructed as specified in Section 600, Overland Park Superpave Asphaltic Concrete Surface and Intermediate Course, of the *Construction Specifications*. If specified asphaltic concrete is not available, the City Engineer, at their discretion, may allow the use of an alternate asphaltic concrete mix.

All concrete used in Right-of-way construction shall be in compliance with the Kansas City Metro Materials Board (hereinafter "KCMMB") requirements that include approved concrete mix designs, including high early concrete mixes.

The provisions of the *Standard Specifications for State Road and Bridge Construction, Kansas Department of Transportation*, current edition and special provisions (hereinafter "*Standard Specifications*"), are incorporated herein by reference, except as hereinafter specified.

All provisions of the *Traffic Control Handbook* for the City of Overland Park are incorporated herein by reference. All provisions of the *Streetlight Design Manual* for the City of Overland Park are incorporated herein by reference.

Failure of the permittee to schedule an inspection by the City Engineer at the start of backfilling operations shall result in re-excavation, removal and recompaction of the backfill, and/or repair pavement, at the permittee's expense, to the reasonable satisfaction of the City Engineer.

1.1. Definitions

Paved Portion of the Right-of-Way is the area within any existing or future public street pavement, under any existing or future curb and gutter, median, asphalt or concrete path, concrete sidewalk, bridges, or existing concrete, asphalt, or gravel driveway approach. It also includes the area under any future driveway approach, when the location is known.

Unpaved Portion of the Right-of-Way is any area outside of the Paved Portion of the Right-of-way.

Clear Zone is the total roadside border area, starting at the edge of the traveled way, available for safe use by errant vehicles.

1.2. Notification of Emergency Services

The Overland Park Police Department, Fire Department and Med-Act shall be notified 24 hours in advance of any street closure. The City Engineer will allow closure of streets only with prior approval at the time the permit is issued.

Overland Park Police Department
(Non-Emergency).....(913) 895-6300

Fire Department and/or MED-ACT Ambulance
(Non-Emergency).....(913) 432-1717

1.3. Contact Information

City Contacts

Lorraine Basalo, Public Works Director/City Engineer.....(913) 895-6023

Kyle Dieckmann, Deputy City Engineer……..(913) 895-6053

Robert Ehlers, Right-of-Way Supervisor……..(913) 895-6189

Eric Keenan, Supervisory Civil Engineer……(913) 895-6028

Utility Contacts

Kansas 811……..(800) 344-7233

Overland Park Traffic Service……...(913) 327-6600

Water One……(913) 895-1800

Johnson County Wastewater…….(913) 681-3200, Ext. 2125

Other Contacts

Automated Information Mapping System (AIMS).....(913) 715-1600

USDA, NRCS - Soil Survey Division……(785) 823-4558

Kansas Dept. of Health and Environment (KDHE).....(785) 842-4600

2. RELOCATION OF SERVICE PROVIDER FACILITIES

2.1. General Requirements

Any necessary relocation of a Service Provider's Facilities shall be subject to: (i) the requirements of OPMC Chapter 13.12 (including but not limited to OPMC 13.12.400); (ii) the provisions set forth herein; and (iii) any other applicable rules, regulations and ordinances of the City. Subject to the applicable provisions below, the Service Provider shall proceed with relocations at due diligence upon notice by the City to begin relocation.

In accordance with OPMC 13.12.430, Service Providers shall remove its Facilities and replace or restore any damage or disturbance caused by the removal at its own expense. The City Engineer may allow underground Facilities or portions thereof to remain in place if the City

Engineer determines that it is in the best interest of public health, safety or welfare to do so and approves of the manner in which such Facilities are left to remain in place. When facilities are abandoned in place, they shall be filled with slurry grout in accordance with the Standard Specification, Section 844.

Facility location markers shall be flush mount (maximum diameter of 8 inches) except when required by Federal regulations.

2.2. Relocation for Public Improvement

The Service Provider shall promptly remove, relocate or adjust any Facilities when necessary for a Public Improvement or when reasonably required by the City by reason of public health, safety or welfare. The Facilities shall be relocated as directed by the City in order to avoid conflict with City construction and improvements, and shall be relocated to areas within: (i) the expanded Right-of-way; (ii) a public utility easement provided for such purposes; (iii) any remaining private easements; or (iv) any remaining portions of such easements not condemned by nor disclaimed to the City. In the event the Service Provider is required to move its Facilities in accordance with this section, any ordinary Right-of-way Permit fee shall be waived.

2.2.1. Project Planning; Notice; and Timing of Relocation

The Service Provider shall participate in any joint planning, construction and advance notification of Right-of-way work, including coordination and consolidation of Street cut work as directed by the City Engineer and shall cooperate with the City and other Service Providers. The intent of the joint planning is for coordination between the City and affected Service Providers in order to determine the best, most efficient, most aesthetic and least obtrusive use of the Right-of-way, consistent with safety, and to minimize traffic and other disruptions, including Street cuts. As necessary, the Service Provider shall furnish maps showing the location of Facilities of the Service Provider within the project boundaries. (See OPMC 13.12.350 & 13.12.360.)

As soon as working drawings are available for Public Improvements which will require the Service Provider to relocate its Facilities, the City shall provide the Service Provider with written notice of relocations and the anticipated bid letting date of said improvement. It is anticipated that such notice should occur at least (180) days in

advance, if not more. The Service Provider shall respond with any conflicts and a proposed construction schedule within (30) days of receipt of the working drawings.

Following notice by the City in the form of the delivery of final design plans for such Public Improvements, and as applicable, notice of acquisition of any necessary City easements or Right-of-way for the relocation, the Service Provider shall remove and relocate its Facilities in accordance with the mutually agreed upon schedule, provided the project is not delayed by adverse weather conditions and other factors beyond the control of the Service Provider. The Service Provider will provide the City with timely notice of any relocation delay and the reasons therefore. The Service Provider shall certify to the City, in writing, that its Facilities have been relocated or adjusted to clear construction in accordance with project plans provided by the City.

It is the intent of this section for both the City and the Service Provider to cooperate with one another so that the need for Facility relocation is minimized and, when required and feasible, relocations may be completed prior to receipt of bids by the City for a Public Improvement.

2.2.2. Facilities located in Right-of-Way

If the Facilities being relocated or adjusted are located in the Right-of-way, the removal, relocation or adjustment shall be performed by the Service Provider at the Service Provider's sole expense without expense to the City, its employees, agents, or authorized contractors. *(See next paragraph if Facilities originally located in a private easement.)*

2.2.3. Facilities location in Private Easement

If the Facilities being relocated are located in a private easement dedicated for the benefit of the Service Provider, or if the Facilities were previously relocated from such a private easement for the benefit of a prior Public Improvement, the removal, relocation or adjustment shall be performed by the Service Provider at the City's expense. The Service Provider shall disclaim those parts of its easements which lie within the expanded Right-of-way. Should the City, in the future, elect to require the Service Provider to again relocate its Facilities to other areas within the Right-of-way or within a public utility easement, the cost of any such future relocation shall be borne by the City.

Notwithstanding the foregoing, the City shall not be responsible for the cost for:

a. Any betterment, improvement or increased capacity of Facilities being relocated. For example, if a 6” pipe needs to be relocated, the City will pay the cost to relocate a 6” pipe in the new location. However, if the Service Provider desires to install a larger 18” pipe in the new location instead of a 6” pipe, the Service Provider shall be responsible for the excess cost difference between installing a 6” pipe versus an 18” pipe.

b. Any relocation or replacement of Facilities that have been abandoned or otherwise removed from use or service because of damage, disrepair or obsolescence.

2.2.4. Facilities location in Public Utility Easement

If the Facilities being relocated are located in a public utility easement, the removal, relocation or adjustment shall be performed as set forth above. The responsibility for the relocation costs shall be subject to the specific language of grant and dedication of the utility easement and any applicable state law.

2.2.5. Delay Damages

Any damages suffered by the City, its agents or its contractors to the extent caused by Service Provider's failure to timely relocate or adjust its Facilities, or failure to properly relocate or adjust such Facilities, shall be borne by the Service Provider.

2.3. Relocation for Private Development/Benefit

In the event that a Service Provider is required to move its Facilities for a private benefit or development (as opposed to a City initiated or funded Public Improvement), the Service Provider shall not bear the cost of the relocation or adjustment to the extent of such private benefit or development, and the Service Provider shall not be obligated to commence the relocation or adjustment of its Facilities until receipt of funds for such relocation or adjustment.

2.4. Relocation/Improvement Initiated by Service Provider

A relocation/improvement to a Facility initiated by a Service Provider shall be borne by the Service Provider and subject to all applicable City Right-of-way regulations and all other applicable rules, regulations and ordinances of the City.

The Service Provider shall initiate any joint planning, construction and advance notification of Right-of-way work, including coordination and consolidation of Street cut work as directed by the City Engineer and shall cooperate with the City and other Service Providers. The intent of

the joint planning is for coordination between the City and affected Service Providers in order to determine the best, most efficient, most aesthetic and least obtrusive use of the Right-of-way, consistent with safety, and to minimize traffic and other disruptions, including Street cuts. As necessary, the Service Provider shall furnish maps showing the location of Facilities of the Service Provider within the project boundaries. (See OPMC 13.12.350 & 13.12.360.)

As soon as working drawings are available for the relocation/improvements, the Service Provider shall provide the City with written notice and the anticipated bid letting date of said relocation/improvement. It is anticipated that such notice should occur at least (180) days in advance, if not more. The City shall respond with any conflicts with the work or proposed construction schedule within (30) days of receipt of the working drawings.

Following notice by the City in the form of the approval of a Right-of-way permit for relocation/improvements, the Service Provider will provide the City with timely notice of any relocation and/or improvements delay and the reasons therefore.

It is the intent of this section for both the City and the Service Provider to cooperate with one another so that the need for the Facility relocation/improvements initiated by the Service Provider is minimized and, when required and feasible, these relocations/improvements will be coordinated with ongoing Public Improvement projects and other ROW-user projects.

3. EXCAVATIONS IN THE UNPAVED PORTION OF THE RIGHT-OF-WAY

3.1. Utility Locates

Prior to commencement of any excavation, the permittee shall identify and locate any buried facilities in accordance with Chapter 13.12 of the OPMC. Permittees shall contact Kansas One-Call at 1-800-DIG-SAFE at least 48 hours prior to commencement of any excavation to allow sufficient time for field locates. The permittee will be required to provide a Kansas One-Call verification number before a Right-of-way permit is issued by the City.

3.2. Utility Locates for City Owned Facilities

The City owns, operates, and maintains the following utility systems within the City: Storm Sewer, Traffic Signals, Fiber Optic Interconnect lines and Street Lights. Emergency utility locate requests may be requested by calling (913) 327-6600 during the hours of 7:30 AM to 4:00 PM. If the emergency locate request is needed after normal business hours or on a weekend,

please call Overland Park Police Dispatch at (913) 895-6300.

3.3. Utility Potholes

Potholes used for the purpose of locating, accessing or repairing buried utilities under pavement shall be accomplished using the methods noted in the Street Repairs included in the current *Standard Details*.

3.4. Coring

Excavation requires coring a circular hole through the entire depth of the pavement using appropriate drilling/coring equipment and removal of the intact pavement core. The vertical alignment of the coring operation shall be perpendicular to the horizon.

Repair of pavement core will be completed in accordance with the Street Repair Detail included in the current Standard Details and the pavement core will be properly disposed of by the permittee outside of the Right-of-way.

3.5. Excavation

Soil shall be removed by air/vacuum extraction methods to expose utilities. The zone of soil removal shall remain essentially within a vertical plane extending below the edges of the core hole. The contractor shall dispose of all extracted and excess materials. Open excavations in unpaved Right-of-way must be secured by temporary fencing that shall be maintained for the duration of the project. In the event there is inactivity on the excavation in excess of 14 days, without cause or reason and notification, the excavation must be returned to as good or better condition.

3.6. Temporary Cover of Core Hole

In the event that a keyhole cored pavement cannot be immediately repaired and core hole will be left unattended, the opening shall be covered with an appropriately sized steel plate fitted with a pilot shaft that is no smaller in diameter than the core hole diameter minus 1” and that extends vertically down into core hole no less than 12”. The plate construction shall prevent the temporary plate from tipping, tilting, bouncing or spinning out of the hole under traffic conditions and shall be capable of supporting normal traffic loads.

3.7. Backfilling

Removable flowable fill, in conformance with Section 843 of the *Construction Specifications*, shall be used for backfill of the core hole. Removable flowable fill shall be placed to the top of the existing subgrade.

3.8. Trenching

The contractor shall not open more trench in advance than is necessary to expedite the work. One (1) block or 400 feet (whichever is the shorter) shall be the maximum length of open trench permitted on any line under construction. All excavations shall be properly protected and fenced.

3.9. Bracing and Shoring

Contractor shall provide adequate bracing, sheeting, and shoring, as necessary, to provide protection for the workmen and the work. All bracing, sheeting, or shoring shall conform to Section 817.3.f, of the *Construction Specifications*. The contractor shall brace and shore all trenches in full accordance with Occupational Safety and Health Standards - Excavations; Final Rule 29 CFR Part 1926.

3.10. Inspection Requirements for General Permits in the Unpaved Portion of the Right-of-Way

For general permit activity, not associated with a new one / two-family residential home building permit, in unpaved portion of the Right-of-way; permittee shall notify the City Engineer to schedule a minimum of one inspection: upon completion of all Right-of-way restoration activities, including concrete, sod or seed work. If weather conditions are such that concrete, sod or seed work cannot be performed, the permittee shall notify the City Engineer after work is substantially complete, except for weather dependent work.

The Permittee shall notify the City Engineer to schedule a second inspection after all restoration work has been completed, including concrete, sod or seed. When all restoration work is completed to the reasonable satisfaction of the City Engineer, the two-year maintenance period will begin.

In addition to the required scheduled inspections, the City Engineer may choose to inspect the on-going permitted work at any time and may require compaction testing at the permittee's expense to ensure that all requirements of the approved permit are being met.

3.11. Inspection Requirements for Permits Associated with a Residential Home Building Permit Unpaved Portion of the Right-of-Way

For permit activity associated with new one/two-family residential home building permits, in the unpaved portion of the Right-of-way; permittee will notify the City Engineer to schedule one inspection, a minimum of 24 hours in advance of the start of backfilling operations. When all backfill work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

In addition to the required scheduled inspections, the City Engineer may choose to inspect the on-going permitted work at any time and may require compaction testing at the permittee's expense to ensure that all requirements of the approved permit are being met.

If the excavation occurs on the opposite side of the street from the permit location, in front of an existing residence; the permittee will be required to restore the Right-of-way to its original condition. This work shall include clean up, grading, and sodding.

Permittee shall notify the City Engineer to schedule a final inspection after all restoration work has been completed. When all restoration work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

3.12. Backfill

Embedment material under and around pipe or conduit shall be as specified by utility Service Provider. If not otherwise specified, embedment material around pipe or conduit shall be crushed stone or siliceous gravel meeting the requirements of Section 1107, of the *Standard Specifications*, Aggregate Designation PB-2.

Embedment material under and around pipe or conduit shall be filled to a maximum of twelve (12) inches above the top of the pipe, except for RCP storm sewer installation. For RCP storm sewer installation, pipe embedment material shall be filled to a maximum of four (4) inches

above the top of the pipe. Removable flowable fill may be used in lieu of embedment material.

Tamped soil backfill may be used in the unpaved portion of the Right-of-way. All unsuitable materials shall be removed and disposed of off site. Tamped soil backfill shall be finely divided job excavated material, free of debris, organic material, and stones, placed in a maximum of eight (8) inch loose lifts, compacted, using vibratory equipment for each lift, to a minimum of 95% standard proctor density. Under no circumstances shall any paving material from excavations in the street be used for backfill material.

Removable flowable fill material, in conformance with Section 843 of the *Construction Specifications*, may be used in lieu of soil backfill**.** Removable flowable fill, when used in excavations in proposed roadways, shall be placed up to the existing surface.

Removable flowable fill, when used outside the roadway, shall be placed up to within 18 inches of the existing surface. Soil, placed in the top 18 inches, shall be generally free of clods, rocks, trash, and other debris and shall be suitable for supporting vegetation.

If an excavation cannot be backfilled and will be left unattended, the contractor shall adequately cover and /or fence the excavation. No excavation shall be left unattended in excess of 72 hours, without permission of the City Engineer.

4. EXCAVATIONS IN THE PAVED PORTION OF THE RIGHT-OF-WAY

4.1. Excavations in Existing Public Streets

4.1.1. Utility Locates

Prior to commencement of any excavation, the permittee shall identify and locate any buried facilities in accordance with Chapter 13.12 of the OPMC. Permittees shall contact Kansas One-Call at 1-800-DIG-SAFE at least 48 hours prior to commencement of any excavation to allow sufficient time for field locates. The permittee will be required to provide a Kansas One-Call verification number before a Right-of-way permit is issued by the City.

4.1.2. Utility Locates for City Owned Facilities

The City owns, operates, and maintains the following utility systems within the City: Storm Sewer, Traffic Signals, Fiber Optic Interconnect lines and Street Lights. Emergency utility locate requests may be requested by calling (913) 327-6600 during the hours of 7:30 AM to 4:00 PM. If the emergency locate request is needed after normal business hours or on a weekend, please call Overland Park Police Dispatch at (913) 895- 6300.

4.1.3. Trenching

The contractor shall not open more trench in advance than is necessary to expedite the work. One (1) block or 400 feet (whichever is the shorter) shall be the maximum length of open trench permitted on any line under construction. Any trench left open overnight shall be properly plated and protected.

4.1.4. Sawing and Benching a Street Cut

Prior to replacement of any type of pavement, the pavement shall be sawed on each side of the required excavation to its full depth.

The size of the saw cut shall be 12 inches greater in all directions than the size of the excavation. This is to provide a minimum of a 12-inch bench of undisturbed subgrade surrounding the excavation. All broken pavement shall be removed from the site.

4.1.5. Sawing Concrete Curb

Concrete curb shall be sawed to its full depth and removed and replaced to the nearest contraction or expansion joint. If an adjacent section of curb and gutter is cracked, chipped, or otherwise damaged in the process of removal, it shall also be removed and replaced to the nearest joint. See also Concrete Construction / Replacement, Section 5 of this document.

4.1.6. Bracing and Shoring

Contractor shall provide adequate bracing, sheeting, and shoring, as necessary, to provide protection for the workmen and the work. All bracing, sheeting, or shoring shall

conform to Section 30.3.f of the *Construction Specifications*. The contractor shall brace and shore all trenches in full accordance with Occupational Safety and Health Standards - Excavations; Final Rule 29 CFR Part 1926.

4.1.7. Inspection Requirements for General Permits Paved Portion of the Right-of-Way

For general permit activity in the paved portion of the Right-of-way, permittee will notify the City Engineer to schedule a minimum of two (2) inspections, an initial inspection and a final inspection. The initial inspection shall occur a minimum of 24 hours in advance of the start of backfilling operations in any cut in an existing street or excavation under future public street; or under any existing or future curb and gutter, median, asphalt path, concrete sidewalk, or driveway approach and a final inspection upon completion of all Right-of Way restoration activities, including concrete, asphalt, sod, or seed.

If weather conditions are such that concrete, asphalt, sod, or seed work cannot be performed, the permittee shall notify the City Engineer after work is substantially complete, except for weather dependent work. The permittee shall notify the City Engineer to schedule a third inspection after all restoration work has been completed, including concrete, asphalt, sod, or seed. When all restoration work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

If permitted activity includes concrete work, such as replacement of curbs, new construction or replacement of sidewalks and/or driveway approaches; an additional inspection shall be required. This inspection shall include form locations and grades, and subgrade prior to the placement of any concrete. Contractor shall notify the City Engineer to schedule an inspection a minimum of 24 hours in advance of concrete placement. Permittee shall notify the City Engineer to schedule a final inspection after all restoration work has been completed, including backfill, Right-of-way grading, clean up and sod.

Except in the event of an emergency, permittee shall notify the City Engineer a minimum of three (3) days in advance of any street closure. No such closure shall take place without notice and prior authorization from the City. See also Notification

Emergency Services Section 1.2 and Traffic Control Section 11.3 of this document.

4.1.8. Inspection Requirements for Permits Associated with a Residential Home Building Permit Paved Portion of the Right-of-Way

For permit activity associated with a new one/two family residential building permit, in the paved portion of the Right-of-way; permittee will notify the City Engineer to schedule one inspection, a minimum of 24 hours in advance of the start of backfilling operations. When all backfill work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

The builder, under the building permit, shall be responsible for constructing the future sidewalk and driveway entrance and complete restoration work, including clean up, grading, and sodding.

If permitted activity includes concrete work, such as replacement of curbs, new construction or replacement of sidewalks and/or driveway approaches; an additional inspection shall be required. This inspection shall include form locations and grades, and subgrade prior to the placement of any concrete. Contractor shall notify the City Engineer to schedule an inspection a minimum of 24 hours in advance of concrete placement. Permittee shall notify the City Engineer to schedule a final inspection after all restoration work has been completed, including backfill, Right-of-way grading, clean up and sod.

If the excavation occurs on the opposite side of the street from the permit location, in front of an existing residence; the permittee will be required to restore the Right-of-way to its original condition. This work shall include any clean up, grading, and sodding. The permittee shall notify the City Engineer to schedule a final inspection after all restoration work has been completed. When all restoration work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

4.1.9. Backfill

Embedment material around pipe or conduit shall be as specified by the applicable Service Provider. If not otherwise specified, embedment material around pipe or conduit

shall be crushed stone or siliceous gravel meeting the requirements of Section 1107 of the *Standard Specifications*, Aggregate Designation PB-2.

Embedment material under and around pipe or conduit shall be filled to a maximum of 12 inches above the top of the pipe, except for RCP storm sewer installation. For RCP storm sewer installation, pipe embedment material shall be filled to a maximum of four (4) inches above the top of the pipe. Removable flowable fill, in conformance with Section 843 of the *Construction Specifications*, may be used in lieu of embedment material.

No soil backfill shall be used in the paved portion of the Right-of-way. Above embedment material, removable flowable fill shall be placed up to the bottom of the surrounding existing pavement.

4.1.10. Compaction Testing

Permittee shall employ a testing laboratory approved by the City Engineer, which shall certify the proper backfilling of any excavation in the paved portion of the Right-of-way. The permittee shall pay all costs associated with such testing. A list of approved testing laboratories shall be on file in the office of the City Engineer. The compaction testing provision shall be waived when removable flowable fill is used as backfill, or with the permission of the City Engineer. The permittee shall provide a copy of the compaction test results to the City Engineer's office prior to the final inspection. Start of the two-year maintenance period shall not commence until the compaction test results have been received and approved by the City Engineer. If test results do not meet specified compaction requirements, the permittee, at his own expense, shall be required to re-excavate, remove and recompact backfill, and repair pavement to the reasonable satisfaction of the City Engineer.

4.1.11. Plating a Street Cut

In accordance with Chapter 13.12 of the OPMC, any excavation left overnight on any thoroughfare or collector street shall be adequately covered with a steel plate. The plate shall be securely anchored, and all edges of the plate shall be ramped with hot mix asphaltic concrete. If cold weather prohibits the availability of hot mix asphaltic concrete, cold mix may be used, so long as it is maintained in a smooth and drivable

condition.

The permittee shall be required to post a WB-23 “Steel Plate Ahead” warning sign with flashing light a minimum of 250 feet ahead. See also the current edition of the *Traffic Control Handbook*. Any excavation left overnight on any residential street shall either be plated as stated above or backfilled up to the surface of the street. Under extenuating circumstances, if an excavation cannot be backfilled, and must be left unattended overnight, the excavation shall be adequately covered. If temporary surfacing material is used, it shall be maintained in a smooth and drivable condition.

No excavation shall be left unattended in excess of 72 hours, without permission of the City Engineer. The permittee assumes the sole responsibility for maintaining proper barricades, plates, safety fencing and/or lights as required from the time of opening of the excavation until the excavation is surfaced and opened for travel.

4.1.12. Repairing a Street Cut

In accordance with Chapter 13.12 of the OPMC, in addition to its own street cuts, Permittee must also restore any area within 5 feet of the new street cut that has been previously excavated, including the paving and its aggregate foundations. In the event of lengthy longitudinal street cuts, the City Engineer may require the entire lane to be repaved.

See Street Repair details of the current *Standard Details* for repair methods..

4.1.13. Permanent Pavement Markings

The permittee shall be responsible for the replacement of permanent pavement markings on thoroughfare or collector type streets, which have been removed or disturbed as a result of any street cut. The permittee will be required to place temporary pavement markings immediately after placing pavement surface, until permanent pavement markings can be installed.

Permanent pavement markings shall be replaced with like materials, in accordance with the latest edition of the "Manual on Uniform Traffic Control Devices", within 14 days after the pavement surface has been placed, unless otherwise authorized by the City

Engineer.

4.2. Excavations for Utility Relocation Work for Publicly Funded Street Projects

Excavations for utility relocation work for Public Works street projects shall conform to the requirements for excavations in existing public streets in the paved portion of the Right-of-way with the following modification: Limits of flowable fill backfill shall extend under the proposed curb and gutter, as well as, under the proposed street pavement.

4.3. Excavations under Existing Concrete Sidewalks or Driveway Entrances

Excavation in areas under existing concrete sidewalks or concrete driveway entrances shall conform to the requirements for excavations in existing public streets in the paved portion of the Right-of-way with the following modification:

4.3.1. Sawing Concrete Sidewalks or Driveway Entrances

Existing concrete sidewalk or concrete driveway entrances shall be sawed to their full depth and removed and replaced to the nearest contraction or expansion joint. If an adjacent section of sidewalk or driveway is cracked, chipped, or otherwise damaged in the process of removal, it shall also be removed and replaced to the nearest joint. See also Concrete Construction / Replacement.

4.4. Excavations under Existing Asphalt Paths

Excavation in areas under existing Asphalt Paths shall conform to the requirements for excavations in existing public streets in the paved portion of the Right-of-Way with the following modification:

4.4.1. Repairing existing Asphalt Paths

Existing asphalt paths shall be sawed to their full depth and removed and replaced for the entire width of the trail. The width of any path repair shall be the same as that being replaced. The minimum allowable thickness of replaced trail material shall be four (4) inches of asphalt and four (4) inches of aggregate base course.

All asphalt paths shall be constructed or replaced in accordance with Section 611 of the

Standard Specifications, Section 1104 of the Construction Specifications and the current Standard Details.

4.5. Excavations in Future Public Residential or Collector Streets

This section applies to the full width of proposed and existing public street Right-of-way where residential or collector streets are not in place at the time of utility installation, but will be constructed at a future date. Excavations in areas where future public residential or collector street pavement will be constructed shall conform to the requirements for excavation in existing public streets, in the paved portion of the Right-of-way, with the following modification:

4.5.1. Backfill for Sanitary Sewer Construction

Allowable backfill materials shall include tamped soil backfill. Tamped soil backfill shall be finely divided job excavated material, free of debris, organic material, and stones, placed in a maximum of eight (8) inch loose lifts, compacted, using vibratory equipment for each lift, to a minimum of 95% standard proctor density. All unsuitable materials shall be removed and disposed of off site.

Tamped soil backfill shall be placed in a maximum of eight (8) inch loose lifts, and shall be compacted, using vibratory equipment for each lift, to a minimum of 95% of standard proctor density. The permittee shall employ a testing laboratory approved by the City, which shall certify the proper backfilling of all trenches under future street pavement. Testing shall be completed for each lift. The permittee shall provide a copy of the compaction test results to the City prior to the final subgrade preparation. If test results do not meet specified compaction requirements, the permittee at his expense, shall be required to re-excavate, remove and re-compact backfill, and repair pavement to the reasonable satisfaction of the City.

5. HORIZONTAL DIRECTIONAL DRILLING

5.1. Utility Locates

Prior to commencement of any excavation, the permittee shall identify and locate any buried facilities in accordance with Chapter 13.12 of the OPMC. Permittees shall contact Kansas

One-Call at 1-800-DIG-SAFE at least 48 hours prior to commencement of any excavation to allow sufficient time for field locates. The permittee will be required to provide a Kansas One-Call verification number before a Right-of-way permit is issued by the City.

5.2. Utility Locates for City Owned Facilities

The City owns, operates, and maintains the following utility systems within the City: Storm Sewer, Traffic Signals, Fiber Optic Interconnect lines and Street Lights. Emergency utility locate requests may be requested by calling (913) 327-6600 during the hours of 7:30 AM to 4:00 PM. If the emergency locate request is needed after normal business hours or on a weekend, please call Overland Park Police Dispatch at (913) 895-6300.

5.3. Utility Potholes

Potholes used for the purpose of locating, accessing or repairing buried utilities shall be as noted below or as necessary to protect existing utilities.

a) Positively identify (by potholing) all crossed utilities that are expected to be as follows:
   - Above and within 5' of the proposed vertical alignment,
   - Below and within 3' of the proposed vertical alignment

b) Positively identify (by potholing) all parallel utilities at the beginning and ending of all bores and as follows:
   - Every 200' if it is within 5' of the proposed alignment,
   - Every 50' if it is within 3' of the proposed alignment

5.4. HDD Construction Safety Guidelines

Prior to performing work involving HDD under a Right-of-way Permit, the Permittee (or its Contractor) shall have a planned response in the event of a utility strike including utility owner notification to avoid the following:

- Electrocution in the event of an electric strike,
- Combustion in the event of a gas line strike,

- Contamination in the case of a sewer strike.

5.5. HDD Drilling Fluid Containment and Disposal Requirements

The HDD Contractor shall contain, handle, and dispose of drilling fluids in accordance with the following requirements:

a) All drilling fluid and fluid additives shall be disclosed, and Material Safety Data Sheets (MSDS) shall be provided to the City Engineer upon request.

b) Excess drilling fluid shall be confined in a containment pit at the entry and exit locations until recycled or removed from the site.

c) Precautions shall be taken to ensure that drilling fluid does not enter roadways, streams, municipal storm or sanitary sewer lines, and/or any other drainage system or body of water.

d) Unintended surfacing of drilling fluid shall be contained at the point of discharge and recycled or removed from the site.

e) Drilling fluids that are not recycled and reused shall be removed from the site and disposed at an approved disposal site.

f) Drilling fluids shall be completely removed from the construction site prior to backfilling or restoring the site.

g) Collection, transportation, and disposal of drilling fluids shall be environmentally safe and comply with local ordinances and government regulations.

5.6. Inspection Requirements for HDD within the Right-of-Way

For general HDD in the Right-of-way, the permittee will notify the City Engineer to schedule a minimum of two (2) inspections, an initial inspection and a final inspection. The initial inspection shall occur a minimum of 24 hours in advance of HDD operations and a final inspection upon completion of all Right-of Way restoration activities, including concrete, asphalt, sod, or seed.

If weather conditions are such that concrete, asphalt, sod, or seed work cannot be performed,

the permittee shall notify the City Engineer after work is substantially complete, except for weather dependent work. The permittee shall notify the City Engineer to schedule a third inspection after all restoration work has been completed, including concrete, asphalt, sod, or seed. When all restoration work is completed to the reasonable satisfaction of the City Engineer, the Right-of-way permit will be closed and the two-year maintenance period will begin.

If permitted activity includes concrete work, such as replacement of curbs, new construction or replacement of sidewalks and/or driveway approaches; an additional inspection shall be required. This inspection shall include form locations and grades, and subgrade prior to the placement of any concrete. Contractor shall notify the City Engineer to schedule an inspection a minimum of 24 hours in advance of concrete placement. Permittee shall notify the City Engineer to schedule a final inspection after all restoration work has been completed, including backfill, Right-of-way grading, clean up and sod.

Except in the event of an emergency, permittee shall notify the City Engineer a minimum of three (3) days in advance of any street closure. No such closure shall take place without notice and prior authorization from the City. See also Notification Emergency Services Section 1.2 and Traffic Control Section 11.3 of this document.

6. CONCRETE CONSTRUCTION / REPLACEMENT

All concrete construction shall meet the requirements of Section 401 of the *Construction Specifications*.

6.1. Materials

6.1.1. Concrete Mix

All concrete used in construction of curbs, sidewalks, sidewalk ramps, and driveway entrances shall be classified as KCMMB 4K having a minimum 28 day compressive strength of 4000 pounds per square inch. Approved concrete mix designs can be found on-line at the Kansas City Metro Materials Board website prior to placement.

6.1.2. Curing Materials

All concrete curbs, sidewalks, sidewalk ramps, or driveway entrances shall be cured either by wet covering, waterproof covering, or liquid membrane seal. The curing period shall be a minimum of 5 days. Curing shall be commenced as soon as possible after the finishing operation and when the concrete has set sufficiently so that it will not be damaged in the process. Concrete curing materials shall conform to Section 401.2.e of the *Construction Specifications*.

6.1.3. Reinforcing Steel

All reinforcing steel bars shall be in conformance with Section 711 of the *Construction Specifications*. All reinforcing bars shall be held in place and positioned by pins or bar chairs. Reinforcing bars for curbs or driveway entrances (if required) shall be new billet ASTM A615 Grade 40.

Welded wire fabric, if used, shall be in conformance with Section 401.2.f. of the *Construction Specifications*, meeting ASTM A815, Grade 60. Welded wire fabric shall be in sheet form. Rolled wire mesh shall not be allowed. Reinforcement of sidewalks will not be required.

6.2. Construction Requirements

6.2.1. Subgrade

All subgrade for curbs, sidewalks, sidewalk ramps, and driveway entrances shall be uniformly compacted and evenly graded to the required subgrade elevation. All loose or extraneous material shall be removed from the subgrade and soft spots shall be uniformly recompacted prior to placement of concrete. The permittee shall have available adequate vibratory compaction equipment to accomplish the compaction as set forth above.

6.2.2. Forms

Forms used in concrete construction shall be of steel or wood, free from warp and shall be sufficiently strong and rigid and securely staked and braced to obtain a finished product correct to the dimensions, lines and grades required. All forms must be cleaned and oiled before each use. A slip-form machine, with electronic controls, may be used

in lieu of forms for long sections of curb or sidewalk.

6.3. Special Weather Conditions

6.3.1. Cold Weather

The Contractor shall comply fully with the provisions of ACI 306.1-90, as modified below:

a. Average daily temperatures as defined in ACI 306.1-90 will be determined and recorded by the City Engineer

b. Concrete temperatures will be determined through the use of high-low thermometers placed and operated by the City below insulated blankets, or where the concrete is uncovered, by checking air temperatures.

c. Uncovered concrete, which has been subjected to freezing temperatures of any duration during the first 24 hours will be considered “frozen,” and shall be rejected.

d. The months of December, January and February will be considered “Cold Weather” and will require concrete protection, regardless of temperature.

e. Concrete shall reach 75% of its design strength prior to backfilling. This strength can be determined through the use of field-cured cylinders, made and tested at contractor’s expense.

f. Concrete must have 5 days where the average daily temperature is above 50 degrees Fahrenheit prior to backfilling unless field cured cylinders are taken. These days do not need to be consecutive.

6.3.2. Hot Weather

Concrete operations in hot weather shall conform to Section 401, Table 401-5 of the *Standard Specifications*.

6.4. Concrete Curb

Concrete curb shall be constructed or removed and replaced in accordance with the requirements of Section 825 of the *Construction Specifications*.

The surface shall be shaped by use of a steel tool to produce the sections shown in the *Standard Details*. The edges shall be rounded with edgers to form the radii as indicated in the *Standard Details*. The surface shall be finished with a wooden or steel float and brushed.

One half (1/2) inch pre-molded bituminous, non-extruding, and resilient expansion joints shall be placed at points of curvature, curb returns, curb inlet transitions, and at intervals not to exceed 250 feet. The material shall extend through the full curb section. After curing, the joints shall be sealed with urethane sealant meeting ASTM C920. The sealant shall be Class 35 (+-35% Joint Movement), Type S and Grade NS and to a depth of one-half (1/2) inch.

Contraction joints shall be two (2) inches deep, and placed at 15 foot intervals. If sawed, the sawing shall begin as soon as the concrete hardens sufficiently to prevent excessive raveling along the saw cut, and shall finish before conditions induce uncontrolled cracks, regardless of the time or weather.

Curbs shall be accurately placed according to the line, grade, and cross section of the existing adjoining curbs. All replacement curbs must have sufficient grade to achieve positive drainage. The City Engineer may require removal and replacement of additional sections of curb in order to achieve positive drainage.

6.5. Concrete Sidewalks, Paths and Sidewalk Ramps

The width of any sidewalk/path repair shall be the same as that being replaced. The minimum allowable thickness shall be four (4) inches for sidewalks and six (6) inches for paths, except within a driveway entrance or sidewalk ramp, where the minimum allowable thickness shall be six (6) inches. The curb section through a sidewalk ramp shall conform to the current *Standard Details*.

All concrete sidewalks and paths shall be constructed or replaced in accordance with Section 824 of the *Construction Specifications* and the current *Standard Details*.

All sidewalk ramps shall be constructed or replaced in accordance with Section 824 of the *Construction Specifications*, and the current *Standard Details*. If sidewalk and curb replacement is required at a street crossing, which has no sidewalk ramp, the permittee shall be required to construct a sidewalk ramp. If an existing sidewalk ramp is damaged or removed,

the permittee shall be required to construct a new sidewalk ramp, in accordance with the current *Standard Details*. All sidewalk ramps shall be constructed with a detectable warning surface as specified in the *Standard Details*, except ramps at non-signalized driveway entrances. Cast Iron Detectable Warning Plates shall be used. See *Approved Materials List*.

The permittee shall provide adequate tools and equipment to produce quality workmanship in placing and finishing concrete. The sidewalk surface finish shall be a coarse texture wood float and broom finish.

The sidewalk surface shall be marked off into nominal squares of dimensions equal to the width of the sidewalk with a maximum distance between joints of 7 feet. A standard joint tool having a width of 1/8 inch and depth of 1/4 of the sidewalk thickness, having a lip radius of 1/8 inch to 1/4 inch shall be used in forming the joints. All contraction joints in the sidewalk shall be tooled or saw cut to match the adjacent joints.

Contraction joints shall be retooled after brooming. Expansion joints shall be constructed at locations where the new sidewalk is longer than 250 feet; sidewalk abuts existing concrete curbs or driveway entrances. Expansion joints shall be formed with 1/2 inch wide bituminous, non-extruding, and resilient filler and shall extend the full depth of the slab.

6.6. Concrete Driveway Entrances

All concrete driveway entrances shall conform to Section 836 of the *Construction Specifications*. Where construction requires the removal and replacement of existing concrete driveway entrances, such removal shall be accomplished by first sawing the existing driveway entrance full depth and removing all material to be replaced. Expansion joints shall be constructed at the back of curb and where the driveway entrance abuts existing sidewalk. Expansion joints shall be formed with 1/2 inch wide bituminous, non-extruding, and resilient filler and shall extend the full depth of the slab. Concrete driveway entrances shall be constructed or replaced to a minimum thickness of six (6) inches. It shall include welded wire fabric or reinforcing bars only if the existing driveway is so reinforced. Concrete driveway entrances shall receive a non-slip finish obtained by a wood float and hair brush or broom applied transverse to the centerline of the driveway. All contraction joints in concrete driveway entrances shall be tooled. Contraction joints shall be tooled after brooming to provide a "picture frame" appearance.

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6.7. Concrete Paver Stone/Brick Medians, Sidewalks, and Driveway Entrances

All concrete paver stones and paver bricks shall conform to Sections 855 and 856 of the *Construction Specifications*. Where construction requires the removal of concrete paver stones in medians, sidewalks, or driveway entrances, replacement stones shall be red cobblestone-style concrete paver stones. Concrete paver stones shall meet the requirements of ASTM C936. Where construction requires the removal of paver bricks in medians, truck aprons, sidewalk, or driveway entrances, replacement bricks shall meet the requirements of ASTM C1272 Type F and match existing colors. A minimum of four (4) inches of KCMMB 4K concrete base, must be placed and sufficiently cured prior to placement of any paver brick. Concrete used in this work shall conform to KCMMB requirements. An approved concrete mix design shall be on line at www.kcmmb.org prior to placement.

7. STREETLIGHTS

All streetlights, including those constructed to support wireless communications facilities, shall comply with the City's *Streetlight Design Manual*.

8. CONSTRUCTION AND REPLACEMENT OF TOWERS, POLES AND RELATED FACILITIES

This Section of the Manual shall not apply to any existing towers or poles within the Right-of-way.

8.1. Towers and Poles

8.1.1. Location

Towers and poles should be placed as far from the roadway as possible. All towers and poles shall be breakaway or maintain a Clear Zone of 8 feet from the face of curb on tangent sections of roadway and a Clear Zone of 12 feet from the face of curb on the outside of horizontal curves at a minimum on roads where curb exists. If curb does not exist, towers and poles shall be placed outside the clear zone in accordance with the latest edition of the American Association of State Highway and Transportation Officials' *Roadside Design Guide*, Table 3-1 in the 2011 edition, or most current version.

8.1.2. Height

The maximum height which may be approved for a tower, pole, or related transmission equipment in the Right-of-way along a thoroughfare, collector or residential street is set forth in OPMC Section 18.395.070 A and in the *Small Cell Aesthetic Standards* (Appendix A).

8.1.3. Design

All towers and poles shall be monopoles or of some other stealth or stealth technology design unless required by the City Engineer to be architecturally compatible to the surrounding development, and shall meet all design requirements in the *Small Cell Aesthetic Standards* (Appendix A). Guy and lattice towers are not allowed. Furthermore, all towers and poles must be designed in compliance with all current applicable technical, safety, and safety-related codes adopted by the City.

New poles shall be round, tapered, and made from 6063 extruded aluminum, with a T6 temper or as approved by the City Engineer. If existing street light poles along the Right-of-way surrounding the proposed pole have been painted, the new pole shall be painted to match said street light poles. This paragraph shall not prohibit the installation of wood poles that replace existing wood poles, or poles installed along existing service lines. However, no new poles shall be installed for new services, service drops or extensions of an existing service.

8.2. Antennas on Towers and Poles

All antennas installed on towers or poles shall be internal or shall be panel antennas of slim-line design be mounted parallel with the tower, or alternatively, an omni-directional antenna may be placed at the top of the tower when it gives the appearance of being a similarly sized extension of the tower. Antenna bridges and platforms are not allowed on towers or poles. Antennas and related visible facilities installed on a tower or alternative structure shall be of materials and color that are consistent with the surrounding elements so as to blend architecturally with said tower or structure. The antennas and related visible facilities shall be of a neutral color that is identical to, or closely compatible with, the color of the tower or

alternative tower structure so as to make the antennas and related facilities as visually unobtrusive as possible.

8.3. Wires Connected to Towers and Poles

To the extent possible, cable or fiber that connects transmission equipment to an equipment box shall be contained inside the pole. If cables or fiber cannot be installed internally, it shall be flush mounted to the pole and covered with a metal, plastic, or similar material cap that matches the color of the pole and is properly secured and maintained by the provider, or cable or fiber shall be enclosed within conduit or a similar cable cover which should be painted to match the pole.

8.4. Backhaul Connected to Towers and Poles

All new wired backhaul shall be installed underground. Providers using backhaul that utilizes wireless technology shall not disturb any trees or vegetation without approval by, or complying with any conditions set by, the City Engineer.

9. MISCELLANEOUS DRIVEWAY CONSTRUCTION /REPLACEMENT

9.1. General Requirements

All asphalt, decorative, or gravel driveways that are damaged or removed shall be reconstructed to the same widths and with the same material that existed prior to Right-of-way work. Gravel driveways may be replaced with asphalt, but no new construction of gravel driveways will be allowed.

All subgrade for driveway entrances shall be uniformly compacted and evenly graded to the required subgrade elevation. All loose or extraneous material shall be removed from the subgrade and soft spots shall be uniformly re-compacted prior to placement of concrete. The permittee shall have available adequate vibratory compaction equipment to accomplish the compaction as set forth above.

9.2. Asphalt Driveway Entrances

Where Right-of-way activities require the removal and replacement of existing asphalt driveway

entrances, such removal shall be accomplished by first sawing the existing driveway full depth and removing all material to be replaced. Asphalt driveway entrances shall be replaced with a minimum of six (6) inches of commercial grade asphaltic concrete meeting the requirements of Section 611 of the *Standard Specifications*, placed in maximum lifts of four (4) inches. In no case shall it be less than that section being replaced. Placing and compaction of the asphalt driveway pavement shall be in accordance with the *Standard Specifications*.

9.3. Decorative Driveway Entrances

Where Right-of-way activities require the removal of existing decorative rock driveway entrances, such removal shall be accomplished by first sawing the existing driveway full depth and removing all material to be replaced.

Additionally, the exposed edge of the decorative rock treatment on the portion of the drive not removed must be protected before and after placement of adjacent concrete. In the event that the edge is damaged, the decorative rock treatment shall be resawed, removed and replaced at a point where a straight line can be obtained at no additional expense to the city. Replacement and the finish may be modified to insure subsequent bonding of the decorative rock treatment. Prior to placement of the surface treatment, the concrete base must have cured for a minimum of 7 days. Aggregates used in the surface treatment must match the existing aggregate in size, shape, gradation, and color, and placement must conform to the procedures used when the original treatment was placed. The permittee should contact the homeowner to ascertain the source of the materials used, and should use duplicates of those materials wherever possible. Any noticeable difference between the existing treatment and the replacement will be grounds for the rejection of the work.

9.4. Gravel Driveway Entrances

As stated above, no new construction of gravel driveway entrances will be allowed.

Existing gravel driveway entrances may be replaced at existing width, but may not be widened. The replacement material shall consist of a minimum of six (6) inches of AB-3, Overland Park Modified, as specified above, which shall be laid watered, manipulated and compacted in lifts not to exceed three (3) inches.

10. RESTORATION

10.1. Suitable Soil

Soil shall generally be free of clods, rocks, trash, and other debris and shall be suitable for supporting vegetation. The area shall be prepared such that sodding may be placed on bare soil. This will consist of cultivating, smoothing, removing of clods, surface stones of one (1) inch in diameter or larger, and weeds in the top six (6) inches of soil,

10.2. Fertilizer

Fertilizer for sod, in accordance with Section 907 of the *Construction Specifications*, shall be of an approved commercial brand composed of a minimum of 25% “Slow Release Nitrogen", 4-1-2 ratio or similar, such as 18-5-9, for Kentucky bluegrass or fescue sod, and 25-5-10 for zoysia sod.

Fertilizer shall conform to the State fertilizer laws, and shall conform to Section 2108 of the *Standard Specifications*. Furnishing and placing fertilizer shall be in accordance with Section 902 of the *Standard Specifications*. Fertilizer shall be uniform in composition, dry and free flowing, and shall be delivered to the site in the original unopened containers, each bearing the manufacturer's guaranteed analysis. Any fertilizer, which becomes caked or otherwise damaged, making it unsuitable for use, will not be accepted. Fertilizer shall be placed prior to seeding at not less than 1 pound of actual phosphorus per 1000 square feet of sodding or seeding area.

10.3. Sodding

Disturbed turf in developed areas shall be sodded in accordance with Section 907 of the *Construction Specifications*. Seeding will be allowed only in undeveloped areas or in median areas on thoroughfare and collector streets with established buffalo grass. Sod shall be replaced with like species. Kentucky bluegrass, turf-type fescue, or zoysia sod shall be used. In the case of mixtures of bluegrass / fescue and zoysia sod, zoysia shall be used.

Sod shall be machine cut at a uniform soil thickness of 5/8 of an inch, plus or minus 1/4 inch, at the time of cutting. Sod shall not be harvested or transplanted when moisture content (excessively dry or wet) will adversely affect its survival. Sod shall be reasonably free of

disease, nematodes, and soil-borne insects. Sod shall also be free of objectionable grassy and broadleaf weeds.

A clean edge shall be established at the outer limits of the area to be sodded, so that good contact can be made between with the ends staggered in a running bond pattern. Each successively laid strip shall be pressed firmly up against the one next to it or up against the edge of the existing turf, to ensure good contact with no overlapping. Sod shall be staked in places where the slope exceeds 3:1. After placing sod, the area shall be tamped with a hand tamp or rolled with a lawn roller half filled with water. Rolling shall be done in a direction perpendicular to the direction in which the sod lengths were laid.

The permittee shall be responsible for watering sod daily or as often as necessary for a minimum of 20 days or until it is firmly rooted and secure in place. Sod shall be sufficiently rooted and growing prior to the restoration inspection and the commencement of the two-year maintenance period.

10.3.1. Sodding Season

Bluegrass or fescue sod may be planted during the periods of March 1 to May 15 and September 1 to November 15. Bluegrass or fescue sod may be planted during the period, November 15 to March 1, when the soil and sod are workable. If bluegrass or fescue sod is planted between November 15 and March 1, the permittee shall maintain the sod until 20 days after the beginning of the spring sod season or until it is sufficiently rooted and growing. Zoysia sod may be planted during the period April 1 to October 15.

10.4. Seeding

Seeding, in accordance with Section 904 of the *Construction Specifications*, will be allowed only in undeveloped areas or in median areas on thoroughfare or collector streets with established buffalo grass. Hydro-seeding may be allowed in specific situations, in lieu of sodding, with permission of the City Engineer.

All seeding materials, bed preparation, and planting shall conform to the applicable requirements of Section 904 of the *Construction Specifications*. All disturbed areas shall be seeded as soon as practical after construction. All areas to be seeded shall be disked,

harrowed, or hand raked to a minimum of two (2) inches to six (6) inches before application of seed. The seedbed should be uniform and well packed. Seed shall be applied with an acceptable seed drill at a depth of 1/2 inch in a uniform manner. Broadcasting and hand raking to a depth of 1/2 inch will only be used on areas where it is impossible to operate a seed drill. The seed shall be covered to a depth of 1/4 to 1/2 inch with a shallow-set spike tooth harrow or other approved methods. After covering, the areas shall be made firm by rolling.

Mulch shall be spread uniformly in a continuous blanket. The mulch shall be anchored in the soil to a depth of two (2) to three (3) inches into the soil surface. Two or more passes may be required to anchor the mulch. No mulch shall be placed unless it can be anchored on the same day.

The seed mixture shall be turf type tall fescue. The rate of application shall be a minimum of 83% pure live seed at 348 pounds of seed per acre.

Buffalo grass seed shall be used in existing median areas on thoroughfare and collector streets with established buffalo grass. The rate of application shall be a minimum of 72% pure live seed at 45 pounds per acre. If buffalo grass seeding is required at other times of the year, temporary seed shall be used, and the areas shall be fertilized and reseeded with buffalo grass during the above time period.

Temporary seeding shall conform to the requirements of Section 902 of the Construction Specifications, the seed mixture shall be annual ryegrass. The rate of application shall be a minimum of 83% of pure live seed at 90 pounds per acre.

Seeding shall be maintained by the permittee until satisfactory growth is established, prior to the restoration inspection and the commencement of the two-year maintenance period.

10.4.1. Seeding Season

Fescue grass seeding season shall be from **February 15 to April 20** and from **August 15 to September 30**. Buffalo grass seeding season shall be from **November 15 to June 1**. Temporary ryegrass seeding season shall be anytime.

10.5. Protection of Trees

All trees and plants shall be protected against injury from construction operations. The permittee shall take extra measures to protect trees, such as erecting barricades or fences around the drip line, and trimming low hanging branches to prevent damage from construction equipment. Trees shall not be endangered by stockpiling excavated material or storing equipment within the drip line of the tree. No backfill material exceeding four (4) inches in depth shall be placed within the drip line area of any tree. When excavation is required within the drip line of any tree, the permittee shall take extra measures to protect as many roots as possible. All roots to be cut or removed shall be cut with a chainsaw, trencher, or other methods that will leave a smooth cut surface. All roots exposed during excavation shall be protected to prevent the roots from drying out by covering the exposed area with canvas or burlap, peat moss, or mulch, and kept damp until the area has been backfilled.

The City Engineer may grant permission by permit to any ROW-user to trim trees upon or overhanging the Right-of-way so as to prevent the branches of such trees from coming in contact with the facilities of the ROW-user. In the event that any ROW-user severely disturbs or damages the health and safety of any tree, the ROW-user will be required to remove and replace the tree with an approved species from the current Approved Street Tree list at the ROW-user's cost.

10.6. Temporary Erosion and Sediment Control

The permittee shall utilize temporary erosion control methods on the project site to prevent mud or debris from entering the roadway or the storm sewer system, and to prevent damage to existing residential yards. Temporary Erosion and Sediment Control shall conform to Sections 901 and 902 of the *Construction Specifications*. The forms of temporary erosion control shall include, but not be limited to sediment fence, temporary seeding, hydro-mulching, erosion control fabrics, and protection of storm drain inlets. Where land disturbance activities have temporarily or permanently ceased on a portion of the project site for over 14 days, the disturbed area shall be stabilized with hydro-mulch or other similarly effective soil stabilizing BMP's. The permit holder must perform self inspections of sediment and erosion control devices weekly.
Records must be kept of all self inspections and be produced upon request by the City inspector. Care shall be taken to avoid damaging erosion and sediment control devices installed by other parties. Where utility work unavoidably damages other installations, the

contractor shall promptly make repairs. Failure to promptly repair damaged BMP's installed by other parties shall be cause for suspension or revocation of the Right-of-way work permit.

10.7. Stormwater Pollution Prevention Best Management Practices

All construction activities shall be performed in accordance with the National Pollution Discharge Elimination System (NPDES) as regulated by the Environmental Protection Agency (EPA), the Kansas Department of Health and Environment (KDHE), and the City of Overland Park.

The Permittee (or its Contractor) shall implement Best Management Practices (BMPs) to ensure that stormwater runoff is not contaminated by sediment caused by land disturbances associated with construction activities in accordance with Sections 901 and 902 of the Construction Specifications and the Standard Details.. The following goals (or "Golden Rules") shall be applied for all Stormwater Pollution Prevention planning:

a) Ensure that sediment controls are in place prior to disturbance, including grubbing and clearing.

b) Maintain sediment controls throughout the construction and restoration processes.

c) Minimize the overall disturbance and soil compaction.

d) Protect disturbed areas throughout the construction process.

e) Prevent stormwater runoff from entering disturbed areas to the maximum extent possible. Stormwater runoff from disturbed areas shall pass through an appropriate sediment control prior to leaving the site.

f) Never intentionally discharge construction contaminants directly into creeks, rivers, ditches, or storm systems.

g) Complete permanent restoration as soon as possible.

h) Provide temporary erosion protection whenever possible. Mulch/hydromulch with or without tackifiers, seed, gravel, geotextiles, or a combination may be applied even if a disturbed area may and/or will be disturbed again or other permanent measures of stabilization are to follow. Spoil piles shall be covered with a tarp and/or contain with a sediment barrier.

i) Contain disturbed sediment on site. Sediment barriers such as silt fences, sand

bags, wattles, rock checks, erosion control blankets/turf reinforcement mats, and/or sediment traps shall be used to contain sediment on the construction site. Existing vegetation may be used as a sediment filter where minimal grades and sheet flow runoff will occur. Protect storm drain inlets down gradient of disturbed areas that are not fully stabilized or where construction will soon be started using barriers such as sediment logs, high flow silt fences, rocks, wattles, gutter buddies, or a combination. Ensure that all sediment barriers are installed and functioning properly for the duration of the disturbance.

j) Avoid causing flooding in roadways and adjacent right-of-way. Do not block existing culverts and storm inlets except as a last resort. Check inlet protection every day to ensure it is functioning and clean if necessary. In addition, sediment traps and filters should be cleaned after every storm event.

11. PUBLIC SAFETY

11.1. General Requirements

If an excavation cannot be backfilled immediately and will be left unattended, the excavation shall be enclosed with good substantial and sufficient barricades and drums equipped with the appropriate type warning lights and orange safety fencing material which is properly secured around the excavation. No excavation shall be left unattended in excess of 72 hours, without permission of the City Engineer.

The Right-of-way construction site shall be kept clean, neat and orderly. Stockpiling of debris and unsuitable materials beyond normal working hours shall not be permitted. Immediately after construction operations have been completed for the day, all equipment, debris, and unsuitable materials shall be completely removed from the site in order to minimize the damage to finished work and inconvenience to the public and adjoining property owners.

11.2. Public Access

In conformance with Chapter 13.12 of the OPMC, whenever a ROW-user shall excavate the full width of any street, sidewalk, alley, driveway approach or other Right-of-way, it shall be its duty to maintain an adequate passage for vehicles and pedestrians across or around the excavation until it is refilled as specified.

If a section of sidewalk is removed as a result of an excavation, temporary aggregate surfacing shall be placed level with the surface of adjacent sidewalk, or sidewalk shall be closed by installing advance warning signs and barricades until the sidewalk can be reconstructed, backfilled, and reopened. Sidewalk shall be reconstructed in as timely a manner as possible. Special emphasis shall be placed on the timely replacement of sidewalks adjacent to schools, public facilities, or commercial areas with high volume pedestrian traffic.

11.3. Traffic Control

When working on City streets, the permittee shall provide adequate and suitable barriers, signs, warning lights, flaggers, and all other equipment necessary to direct and reroute traffic and protect the public from moving or stationary vehicles, equipment, and materials, and other obstructions. Also, adequate protective warning lights and signs shall be provided to warn of any obstruction or excavation in the street, sidewalk, or parkway. All barricades, signs, and lights shall conform to the latest edition of the *Manual on Uniform Traffic Control Devices*, and the current edition of the *Overland Park Traffic Control Handbook for Street Maintenance and Construction Operations*.

Non-emergency work on thoroughfare or collector streets may not be accomplished during the hours of 7:00 to 8:30 A.M. and 4:00 to 6:00 P.M.

The permittee shall be required to post a WB-24 “Steel Plate Ahead” advance warning sign with flashing light a minimum of 250 feet ahead of a steel plate. See also Plating a Street Cut section 4.1.11 of this document.

All workers in construction zones shall wear a shirt, vest, or jacket that is orange, yellow, strong yellow green or fluorescent versions of these colors. For nighttime work, similar outside garments shall be retroreflective. The retroreflective material shall be orange, yellow, white, silver, strong yellow green or a fluorescent version of one of these colors and shall be visible at a minimum distance of one thousand feet. Flaggers shall be equipped with sign paddles.

A detailed traffic control plan shall be required prior to issuance of permit, for any lane closure on a thoroughfare or collector street. All personnel involved in the preparation of construction traffic control plans and the installation of all traffic control devices shall be certified by The American Traffic Safety Services Association (ATSSA) as a "Traffic Control Supervisor" or by the International Municipal Signal Association (IMSA) as a "Work Zone Safety Specialist" or

other equivalent certification for the design and installation of construction traffic control devices. The certification must be current and readily available for verification at any time as requested by the City Engineer. Minimum requirements for the certification shall include attending training sessions and successful completion of a written examination proctored by a nationally recognized and qualified agency. The permittee shall be required to notify emergency services in the event of any street closure. See also Notification of Emergency Services.

11.4. Facility Separation Requirements

The following are minimum vertical and horizontal separation requirements for facilities in the Right-of-way. The City Engineer reserves the right to modify or require higher standards if deemed necessary to protect the public's health, safety or welfare. The City may also enforce additional standards promulgated by the Kansas Department of Health and Environment, the Kansas Corporation Commission, and within the National Electrical Standards Code.

11.4.1. Water/Wastewater

The minimum separation between any water service or wastewater line and other facility shall be three (3) feet, and a minimum of ten (10) feet of separation from any parallel water or wastewater line, unless otherwise approved by the applicable water/wastewater authority.

11.4.2. Gas

Natural gas lines are subject to DOT 192.325, Federal Pipeline Safety Standards. The minimum separation distance from any existing gas main will be thirty-six (36) inches horizontally, and twelve (12) inches vertically. Lines must cross at a ninety (90) degree angle.

11.4.3. Telephone, Cable, Fiber, and Power

All telephone, cable, and fiber lines shall be installed at a minimum depth of twenty-four (24) inches under unpaved surfaces, and forty-eight (48) inches under paved surfaces. Required separation distances shall be twenty-four (24) inches from natural gas lines, power lines, or other nearest Service Provider. Overhead utilities shall not be installed

or maintained directly in front of or otherwise in a manner which impairs City traffic signals or signs or other warning devices.

11.5. Depth of Facilities

A minimum depth of forty-eight (48) inches shall be maintained from the street surface to the top of any boring. No facilities shall be installed less than forty-eight (48) inches below the street surface of any Right-of-way. The minimum depth for any facilities installed underground outside the street surface is twenty-four (24) inches; provided that underground telecommunications or cable drop lines are exempt from this requirement, but shall be buried at a reasonably safe depth not less than six (6) inches. Any facilities installed less than forty-eight (48) inches under the street surface or thirty-six (36) inches outside of the street surface is done so at the risk of the facility user, and the City shall not be responsible for any damage to the same.

APPENDIX A

Small Cell Aesthetic Standards

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Revised March 1, 2024

SMALL CELL AESTHETIC STANDARDS[1]

The following Small Cell Aesthetic Standards (the "Aesthetic Standards") reflect the desire of the City of Overland Park ("City") to maintain the aesthetics within the City, while allowing for an increase in the availability and quality of wireless services.

These Aesthetic Standards apply to all small cell antenna applications for placement of new small cell antennas on city-owned and non-city-owned poles in the Right-of-way, whether collocated streetlight poles, monopoles or utility poles. Applications that do not conform to these Aesthetic Standards will be denied by the City.

For small cell antenna structures installed on any pole within the Right-of-way, the intent of these Aesthetic Standards is:

1. To establish a clear, defined aesthetic standard for use throughout the City.
2. To minimize unnecessary quantities of new poles by encouraging collocation of small cell facilities.
3. To require, in situations where new poles will be placed, that equipment be placed on new, pre-designed and approved poles such that as much equipment as reasonably possible, including any wiring, can be concealed inside the pole.
4. To require, in situations where attachments will be made to existing poles, that equipment, cabling, and conduit be concealed internally or through the use of approved shrouding or camouflaging equipment.
5. To require that all electrical or control cabinets be located in a manner as to not detract from the aesthetic appeal of adjacent houses, buildings, etc. or to be located as to cause an obstruction to visibility, and be landscaped to provide necessary screening in accordance with the Unified Development Ordinance.

Section 1. Application Requirements

The City may develop new or additional permit application forms, checklists, updated or amended Aesthetic Standards, and other related materials as required to optimally meet the goals of the City, its citizens, and its leadership.

1.1 Site Plans and Structural Calculations

The applicant must submit fully-dimensioned site plans, elevation drawings and structural calculations prepared, sealed, stamped and signed by a Professional Engineer licensed and registered by the State of Kansas. Drawings must depict improvements and the proposed facility, with all proposed transmission equipment, power source, electrical

[1] Intended to include DAS antennas and other similar wireless communications facilities.

service pedestal and other associated access or utility easements and setbacks.

All equipment depicted on the plans shall include:

1. Manufacturer's name and model number
2. Physical dimensions including, without limitation, height, width, depth, volume and weight with mounts and other necessary hardware, and effective projected area (EPA)
3. Technical rendering of all external components, including enclosures and all attachment hardware, including a depiction of how much external wiring will exist.

Section 2. General Design and Construction Standards

The City desires to promote safe, cleanly organized and aesthetically acceptable facilities using the smallest and least obtrusive means available to provide wireless services to the community. All wireless facilities in the Right-of-way must comply with all applicable provisions in these Aesthetic Standards. If any other law, regulation or code requires any more restrictive structural design and/or construction requirements, the most restrictive requirement will control.

Document Superscript Key

The following superscript key shall be used to indicate which various type(s) of small cell installations are governed by the specific design and construction standard listed below:

C - applies to equipment collocated on City streetlight poles

M - applies to monopoles

U - applies to wood or steel utility poles

2.1. RF Cutoff Switch [C,M,U]

All facilities shall be designed, constructed, operated and maintained in compliance with all generally applicable health and safety standards, regulations, and laws, including without limitation all applicable federal regulations for human exposure to RF emissions. The small cell provider shall provide an RF cutoff switch a maximum of 10’ from the finished ground surface at the pole location that is easily reached by maintenance personnel. An RF warning sign shall also be placed on the pole below the cutoff switch.

2.2. Small Cell Antenna [C,M,U]

The small cell antenna shall either be mounted internal to the pole, or top-mounted and concealed within a radome that also conceals the cable connections, antenna mount and other hardware. Any radome, shield or shroud shall meet the following requirements for concealing exposed cable and finish.

2.2.1. Antenna Shroud Requirements

A screening shroud shall be provided on the underside of the small cell antenna, mounted external to the pole, to conceal cable connections from public view. The shroud shall be firmly attached and sealed to prevent birds from entering and nesting.

2.2.2. Finish Requirements

The equipment shroud must be non-reflective and painted or color impregnated to match the color of the existing pole as close as possible.

2.3. Electrical Meter and Cabinet Requirements [C,M]

The electrical meter shall not be installed on the pole. Any necessary meter or other accessory cabinet shall be installed on the outside edges of the street, behind the sidewalk, bicycle or multi-use trail, and said cabinet shall meet all location and landscaping requirements of the City's Unified Development Ordinance. (Requires screening for cabinets with an actual or projected footprint greater than 2 square feet.) The provider shall be required to maintain any required vegetative landscaping to ensure a neat appearance and to mitigate sight distance obstructions. When the installation occurs in an area where the adjacent poles are painted, the City may require that the electrical meter cabinet be painted to match the color of the poles.

2.4. Strand Mounted Small Cell Facilities [U]

Aerial fiber and power strand installations are allowed. However, coiling of excess fiber or other cables is not allowed. All lines shall be neatly trained and secured.

2.4.1. Size Requirements

Any strand mounted cell facility shall not be larger in dimension than 24" in length, 15" in width, and 12" in height , and any exterior antenna is no longer than 11", that are strung on cables between existing utility poles, in compliance with the National Electrical Safety Code and shall be subject to the structural limitations of the Service Provider.

2.4.2. Finish Requirements

The equipment shroud must be non-reflective and painted or color impregnated to match the color of the existing pole, or surrounding infrastructure as close as possible.

2.5. Pole Requirements When Located Within the Clear Zone or Center Median [C,M]

Poles located within the clear zone or center median of any street shall meet the following requirements for design, material, shape, height, diameter and finish. Any collocated poles shall also meet the City standard streetlight details in regard to handhole size and location, standard bolt patterns for luminaire arm attachments, cable hooks, grounding lugs, cabling

access, etc. as required to accommodate and maintain the City infrastructure.

2.5.1. Breakaway Requirements

The following breakaway requirements shall be maintained.

2.5.1.1. Pole Requirements

All poles within the center median of any public street or those poles located on the outside of curb lines within the established clear zone requirements, in accordance with the latest edition of the *City of Overland Park Manual of Infrastructure Standards*, shall be breakaway according to National Cooperative Highway Research Program (NCHRP) 350 or Manual for Assessing Safety Hardware (MASH), latest edition, using approved breakaway couplings or frangible bases. The weight of a small cell pole, including all attached equipment, shall not exceed the total weight as recommended by either the pole manufacturer or manufacturer of the breakaway device. The breakaway pole device shall not exceed 12” in height.

2.5.1.2. Cable Requirements

When poles are required to meet breakaway requirements, all cabling within the pole shall also be required to meet breakaway requirements with appropriate “pull apart” electrical connectors so the cables do not “snag” upon impact by an errant vehicle.

2.5.2. Pole Foundation Requirements

The foundation shall remain essentially flush with the ground so that the breakaway device leaves no more than a 4” stub height above the ground, consistent with AASHTO requirements. The foundation shall be either a cast-in-place reinforced concrete foundation or screw-in foundation meeting the structural requirements of the loaded pole. All foundations shall be accompanied with a detail or shop drawing that is sealed by a Professional Engineer approving of the design.

2.5.3. Pole Design Requirements

The pole shall be designed in accordance with the *2013 American Association of State Highway Transportation Officials (AASHTO) Standard Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals*, or latest version adopted by the City.

2.5.4. Pole Material

The poles shall be manufactured from 6063 extruded aluminum with a T6 temper.

2.5.5. Pole Shape

The cross section of the pole shall be round and shall be fabricated in a

continuous true taper from 2’ from the base to the top of the shaft.

2.5.6. Pole Height

The pole height shall be consistent with the street classification and adjacent poles. The maximum height from the finished ground surface to the top of the antenna shall not exceed the values indicated in the table below:

Maximum Pole Height Requirements Based on Street Classification		
Street Classification	**Street Subcategory**	**Proposed Pole Height to Top of Antenna**
Local	Single Family	30’-0”
	Apartment	40’-0” [1]
	Commercial	40’-0” [1]
	Industrial	40’-0”
Collector	Lite	40’-0”[1]
	Standard	40’-0”[1]
	Super	40’-0”[1]
Thoroughfare	Standard	50’-0”[1]

[1] Maximum allowable height may be more restrictive based on the height of existing streetlights. If existing poles are 20’ or less, the maximum allowable pole height (including antenna) is 30’.

2.5.7. Pole Diameter

The pole diameter measured at the base of the pole shall not exceed the values indicated in the table below:

Maximum Pole Diameter Requirements Based on Proposed Pole Height to Top of Antenna	
Proposed Pole Height to Top of Antenna	**Maximum Outside Diameter at Pole Base**
30’-0”	8”
40’-0”	10”
50’-0”	12”

2.5.8. Pole Finish

Aluminum poles shall have a satin ground finish unless otherwise specified. If adjacent poles are painted, the proposed pole shall be finished meeting the same color chip as the adjacent poles. Aluminum poles that are painted to match adjacent poles shall be painted with a polyester powder coat. (Refer to Section 3.3)

2.6. Pole Requirements When Located Outside the Clear Zone or Center Median [M]

It is presumed that this section will only pertain to monopoles. The poles shall meet the following requirements for design, material, shape, height, diameter and finish.

2.6.1. Pole Design Requirements

The pole shall be designed in accordance with the *2013 American Association of State Highway Transportation Officials (AASHTO) Standard Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals*, or latest version adopted by the City.

2.6.2. Breakaway Requirements

Poles that are located on the outside curb lines beyond the clear zone boundary as established in the latest edition of the City of Overland Park Manual of Infrastructure Standards, are not required to be breakaway.

2.6.3. Pole Material

The poles shall be manufactured from 6063 extruded aluminum with a T6 temper.

2.6.4. Pole Shape

The cross section of the pole shall be round and may be fabricated in a continuous true taper from at least 2’ from the base to the top of the shaft or straight without a taper.

2.6.5. Pole Height

The pole height shall be consistent with the street classification and adjacent poles. The maximum height from the finished ground surface to the top of the antenna shall not exceed the values indicated in the table below:

Maximum Pole Height Requirements Based on Street Classification		
Street Classification	**Street Subcategory**	**Proposed Pole Height to Top of Antenna**
Local	Single Family	30'-0"
	Apartment	40'-0" [1]
	Commercial	40'-0" [1]
	Industrial	40'-0"
Collector	Lite	40'-0"[1]
	Standard	40'-0"[1]
	Super	40'-0"[1]
Thoroughfare	Standard	50'-0"[2]
[1]Maximum allowable height may be more restrictive based on the height of existing streetlights. If existing poles are 20' or less, the maximum allowable pole height (including antenna) is 30'.		

2.6.6. Pole Diameter

The pole diameter measured at the base of the pole shall not exceed the values indicated in the table below:

Maximum Pole Diameter Requirements Based on Proposed Pole Height to Top of Antenna	
Proposed Pole Height to Top of Antenna	**Maximum Outside Diameter at Pole Base**
30'-0"	8"
40'-0"	10"
50'-0"	12"

2.6.7. Pole Finish

Aluminum poles shall have a satin ground finish unless otherwise specified. If adjacent poles are painted, the proposed pole shall be finished meeting the same color chip as the adjacent poles. Aluminum poles that are painted to match adjacent poles shall be painted with a polyester powder coat.

2.7. Utility Pole Requirements [U]

At the approval of the local utility Service Provider, small cell equipment may be installed on wood or steel utility poles as long as they meet the clearance requirements to power lines or other requirements or regulations of the local utility Service Provider.

2.7.1. Antenna and Utility Pole Height

The maximum height from the finished ground surface to the top of the antenna mounted on a utility pole shall not exceed the greater of 10% of the existing utility pole height or 10 feet. In the event a new utility pole is being installed within an existing utility line, the height calculation shall be based upon the immediately adjoining utility poles within the utility line. (This is intended to be a one-time height increase allowance. If multiple height increases are made, they should not cumulatively exceed 10' higher than the original pole height.)

2.8. Luminaire Arm Requirements [C]

The luminaire arm(s) shall meet the following requirements for design, material, shape, length, location and finish according to the City's standard streetlighting details.

2.8.1. Design Requirements

The luminaire arm(s) shall be designed in accordance with the *2013 American Association of State Highway Transportation Officials (AASHTO) Standard Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals*, or latest version adopted by the City.

2.8.2. Luminaire Arm Material

The luminaire arm(s) for 30' and 40' luminaire mounting heights shall be tubing/piping manufactured from 6063 aluminum tubing with a T6 temper. The luminaire arm material shall be the same as the pole material.

The luminaire arm for 14' luminaire mounting heights shall be an aluminum casting as detailed by the City, that is compatible with the residential fixture mounting.

2.8.3. Luminaire Arm Shape

The luminaire arm(s) for 30' and 40' luminaire mounting heights shall match the style and shape as the luminaire arm(s) on the existing pole that is being replaced. The style shall either be a single member arm type or truss type arm. The luminaire arm for a 14' luminaire mounting height shall be rectangular in shape with decorative enhancements as depicted in the special details.

2.8.3.1. Single Member Arm

The single member arm shall be tapered tubing. After tapering, the member shall be flattened to produce an elliptical cross-section with the

major diameter in the vertical plane, perpendicular to the wind. The outboard end of the arm shall remain round with a 2" slipfitter for mounting the luminaire.

2.8.3.2. Truss Type Arm

The truss type member arm assembly shall be a one piece welded assembly consisting of an upper arm and lower arm (brace) securely joined by a vertical strut and a connector or weld at the outboard end of the arm assembly. The upper arm shall be tapered. After tapering, the upper arm shall then be flattened to produce an elliptical cross-section with the major diameter in the horizontal plane, parallel to the wind. The outboard end of the upper arm shall remain round with a 2" slipfitter for mounting the luminaire. The outboard end of the lower arm (brace) shall be covered by an end cap.

2.8.3.3. 14’ Pole Arm Adapter

The luminaire arm for 14’ residential poles shall be a special fabricated mounting bracket to mount the luminaire on the side of the pole instead of at the top of the pole. The design of the arm shall be based on drawing number 19-119-PR101 with bracket arm number SAL-WSB-PB which is on record at the City.

2.8.4. Luminaire Arm Length

The length of the luminaire arm(s) for 30’ and 40’ luminaire mounting heights shall be the same as those on the existing pole that is being replaced. The luminaire arm length for 14’ luminaire mounting heights shall be approximately 17” from the side of pole to the center of the luminaire.

2.8.5. Luminaire Arm Location

The luminaire arm mounting location for 30’ and 40’ luminaire mounting heights shall match the location of the existing poles, such that the mounting height of the proposed luminaire housing matches the mounting height of the adjacent poles and luminaires.

If a collector street currently has 14’ poles, any new pole meeting the requirements for antenna mounting heights may be used. The post top style luminaire shall be mounted at 14’ with the special rivnut mounting hardware and decorative arm. An additional simplex luminaire arm mounting bracket shall be mounted at the top of the pole to accommodate a future 6’ or 8’ single member luminaire arm for 30’ mounting height. The hole in the simplex mounting bracket shall be covered with an aluminum plate and secured with screws.

The luminaire arm for 14’ luminaire mounting heights shall require special rivnut mounting hardware to accommodate the decorative arm.

2.8.6. Luminaire Arm Finish

The arm(s) for 30' and 40' luminaire mounting heights shall be finished the same color of the poles. See section 2.5.6. The arm for the 14' luminaire mounting height shall be painted black to match the luminaire housing.

2.9. Luminaire Requirements [c]

The luminaire(s) shall meet the following requirements for style and finish:

2.9.1. Luminaire Style

The luminaire(s) for 30' and 40' mounting heights shall be LED cobrahead style and be from the City Approved Materials List, which can be found at the link below.

http://ppm.opkansas.org/wiki/index.php/Preapproved_Materials_Street_Lighting#LED_Cobrahead_Luminaire.

The class of luminaire to be used shall be approved by the City. Existing LED cobrahead luminaire(s) shall be replaced with the same class of LED luminaire(s) that already exists on the pole that is being removed. Existing HPS cobrahead luminaire(s) shall be replaced with LED cobrahead luminaire(s) according to the table below:

Luminaire Requirements Based on Existing HPS Luminaire	
Existing HPS Cobrahead Luminaire Wattage	**Required LED Cobrahead Class Luminaire**
400 Watt	Class A
310 Watt	Class B
250 Watt	Class C
150 Watt	Class D
100 Watt	Class E

The luminaire for the 14' post top poles shall be the standard post top fixture as indicated in the City Approved Materials List. It shall be equipped with an LED lamp. All existing luminaire(s) shall be returned to the City of Overland Park Traffic Maintenance Division.

2.9.2. Luminaire Finish

The luminaire(s) shall have a factory applied finish that matches the same color

chip as the luminaires on adjacent poles.

2.10. Cabling Requirements [C,M]

All cabling shall primarily be internal to the pole. Any exposed cabling, external to the pole, shall be minimized. No cable shall be visible at the top of the pole near the antenna. External cables powering the pole mounted radios, cutoff switches or other devices shall be limited to a total length of 24", including drip loops, slack, etc. Any cable access point on the pole shall be sealed with a manufactured product to keep birds from entering and nesting. Duct seal or putty is not an approved product.

Section 3. General Location Criteria

Any new pole and/or equipment and other improvements associated with a new pole or an existing pole must meet the following criteria:

3.1 Pole Location

3.1.1 Collocation

The City desires and encourages collocations between limited numbers of multiple, separate wireless service providers on the same support structure whenever feasible. However, collocated poles shall not be used to replace streetlight poles the City has purchased from Evergy (formerly KCP&L).

3.1.2 General

These general principles shall apply:

3.1.2.1 Monopoles Located in a Raised Street Median

Monopoles, located in a raised street median, shall be placed equidistant between two existing streetlight poles and roughly centered in the median. (See Section 2.5.1 for applicable breakaway requirements.)

3.1.2.2 Monopoles Located on the Outside of Curb Line

Monopoles, located on the outside of the curb line, shall be placed, to the extent possible, approximately equidistant between two existing streetlight poles with preference to being located on the property line between lots and no closer than 3' from the back of curb to the center of the pole. (See Section 2.5.1 for applicable breakaway requirements.)

3.2 Not on Traffic Signal Poles

Small cell equipment shall not be allowed on existing or proposed traffic signal poles or existing or proposed streetlight poles that have existing or proposed traffic signal equipment mounted to them.

3.3 Not on Decorative Poles

Small cell equipment shall not be allowed to collocate on decorative streetlight poles or poles that have decorative luminaires. The term “decorative” shall be applied to anything that does not conform to the City’s Streetlighting *Standard Details*.

3.4 Not to Cause Obstructions

Any new pole and/or equipment and other improvements associated with a new pole, electrical meter cabinet or an existing pole must not obstruct:

1. Any intersection sight distances as required by the American Association of State and Highway Transportation Officials (AASHTO), latest edition adopted by the City.

2. Any above-ground or underground infrastructure for traffic control, streetlight or public transportation, including without limitation any curb control sign, vehicular traffic sign or signal, pedestrian traffic sign or signal, barricade or traffic control equipment.

3. Access to any public transportation vehicles, shelters, street furniture or other improvements at any public transportation stop (including, without limitation, bus stops, bike share stations, etc.

4. Any access to sidewalks, pedestrian facilities, etc. as outlined in the United States Access Board Public Rights of Way Access Guidelines (PROWAG).

5. Access to above-ground or underground infrastructure owned or operated by any public or private utility Service Provider, including fire hydrants, etc.

6. Access to any doors, gates, sidewalk doors, passage doors, stoops or other ingress and egress points to any building appurtenant to the Right-of-way, or access to any fire escape.

Section 4. City Pre-Approved Pole Designs

The City Engineer may require providers to design their pole(s) utilizing pole designs pre-approved by the City Engineer.

Section 5. Amendment

The requirements set forth in these Aesthetic Standards may be amended from time to time by the City Engineer at the City Engineer's discretion. Any amendment of these Aesthetic Standards shall be published once in the official City newspaper and shall become effective upon publication.

Section 6. Exceptions

The City Engineer, in the City Engineer's sole discretion, may grant exceptions to these Aesthetic Standards if the City Engineer's finds the following conditions exist:

1. The Aesthetic Standards as applied to a specific set of circumstances are: (a) technically infeasible and (b) unreasonable when balanced against the interest of avoiding or remedying the intangible public harm of unsightly or out-of-character facility deployments; and

2. The applicant's proposed design and aesthetic appearance for the proposed facilities meets the spirit of these Aesthetic Standards.

Section 7. Design and Construction Standards Manual

The Aesthetic Standards set forth herein regarding streetlight poles are meant to be read in conjunction with the City's Design and Construction Standards Manual criteria for streetlight poles and any applicable provisions in the Unified Development Ordinance. These Aesthetic Standards will be amended as necessary so as to not conflict with the current version of the City's Design and Construction Standards Manual requirements regarding streetlight poles.

Section 8. Severability

The provisions of any part of these Aesthetic Standards are severable. If any provision or subsection, or the application of any provision or subsection to any person, entity or circumstance is held invalid, the remaining provisions, subsections and applications of such Aesthetic Standards to other persons, entities or circumstances shall not be made invalid as well. It is declared to be the intent of this section that the remaining provisions would have been adopted had such invalid provisions not been included in these Aesthetic Standards when originally adopted.

Section 9. Process For Decommissioning Of Small Cell Pole Collocations

The following are three (3) options for small cell companies in the event they desire to decommission a small cell installation:

1. Partial Replacement for Standard Foundations

This option assumes the bolt circle for the small cell pole foundation matches the diameter of a standard sized City streetlight pole. This would require the removal of all existing equipment as noted below and replacement of equipment to bring it in conformance with the condition immediately prior to the small cell installation. The following work is required:

a. Remove all small cell antennas, antenna relays, RRU's, RF Disconnects, etc and all associated cable within the small cell pole.

b. Remove the existing small cell pole and luminaire arm(s). Small cell pole and luminaire arm(s) will become the property of the small cell provider, if desired.

c. The concrete foundation or screw-in foundation that conforms to the bolt circle of a standard City streetlight pole can be left in place and used to install a standard City streetlight pole.

d. Remove any fiber optic or small cell service boxes adjacent to the small cell pole or meter pedestal location.

e. Remove meter pedestal and disconnect electrical service.

f. Remove all electrical cable and/or fiber optic cable within any conduit installed for the small cell installation, including electrical service cable to the Evergy power source and the 1c#10 AWG cable for the streetlights.

g. Remove all LED luminaires from the existing small cell pole. Salvage for reinstallation on the new streetlight pole.

h. Install a new City standard streetlight pole and luminaire arm(s). The streetlight pole and the luminaire arm(s) will be provided by the City.

i. Reinstall LED luminaires salvaged from the small cell pole onto the new streetlight pole.

j. Install new #4 AWG streetlighting distribution cable from the existing Type 2 junction box to the streetlight pole so it extends from 1" to 3" above the top of the handhole opening. Splice the #4 AWG cables to existing #4 AWG cables in the Type 2 junction box.

k. Install new 1c#10 AWG pole and bracket cable for luminaires and ground cable inside the new streetlight pole with 12" to 14" of slack at the handhole, as indicated in the standard details.

l. Install new multi-tap electrical connectors and breakaway fused and unfused electrical disconnects in the base of the new streetlight pole.

m. Remove any traffic signs that were mounted on the small cell pole and reinstall on the new streetlight pole.

n. Restore all disturbed earth and sod.

2. Full Replacement for Non-Standard Foundations

This option assumes the bolt circle for the small cell pole foundation does not match the diameter that would support a standard sized City streetlight pole. This also includes any small cell pole that replaced a former 15' post top pole. The work required, in addition to what is required in Option 1 is as follows:

a. Remove any concrete foundation a minimum of 24" below finished grade or remove the small cell pole screw-in foundation.

b. Install a new City standard screw-in foundation at a different location as approved by the City of Overland Park.

c. Install new 2" HDPE (Gray) conduit from the existing Type 2 junction box to the new streetlight pole location.

3. Escrow Deposit for Future Replacement

In lieu of either of the above options, the small cell provider may opt to place money in escrow for future replacement of the poles to the standard City equipment. Only 30' and 40' poles are eligible for Escrow Deposit for Future Replacement. A small cell pole that replaced a former 15' post top pole is not eligible and will require decommissioning in accordance with Option 2, above.

The small cell provider shall deposit money in an escrow account to cover the out-of-pocket costs the City would incur to replace the equipment with standardized equipment. The cost is based on whether the small cell pole foundation bolt circle matches the diameter of a standard sized City streetlight pole or not. Cost estimates are based on current year costs as determined by the City Engineer . For example in 2022, the following costs would have applied at the time of decommissioning:

a. <u>Costs if the small cell pole foundation can be reused as of 2022</u>:
   - $3,800 for the decommissioning of a 40' twin small cell installation
   - $3,400 for the decommissioning of a 30' single small cell installation

b. <u>Costs if the small cell pole foundation cannot be reused as of 2022</u>:
   - $6,200 for the decommissioning of a 40' twin small cell installation
   - $5,800 for the decommissioning of a 30' single small cell installation
   - $4,800 for the decommissioning of a formerly 14' residential pole

The following work by the small cell provider is still a requirement, in addition to placing money in an escrow account:

a. Remove all small cell antennas, antenna relays, RRU's, RF Disconnects, etc and all associated cable within the small cell pole.

b. Seal all handholes or holes in the pole serving small cell equipment using standard handhole covers, if applicable, or stainless steel or aluminum plates.

c. Remove any fiber optic or small cell service boxes adjacent to the small cell pole or meter pedestal location.

d. Remove meter pedestal and disconnect electrical service.

e. Remove all cable within any conduit installed for the small cell installation, including electrical service cable to the Evergy power source.

f. Install new #4 AWG streetlighting cable from the existing Type 2 junction box to the existing pole location so it extends from 1" to 3" above the top of the handhole opening.

g. Install new multi-tap electrical connectors in the base of the existing pole for electrical connections from #4 AWG to #10 AWG cable.

h. Shorten and reconnect the 1c#10 AWG pole and bracket cable to provide 12" to 14" of slack inside the pole.

i. Restore all disturbed earth and sod.